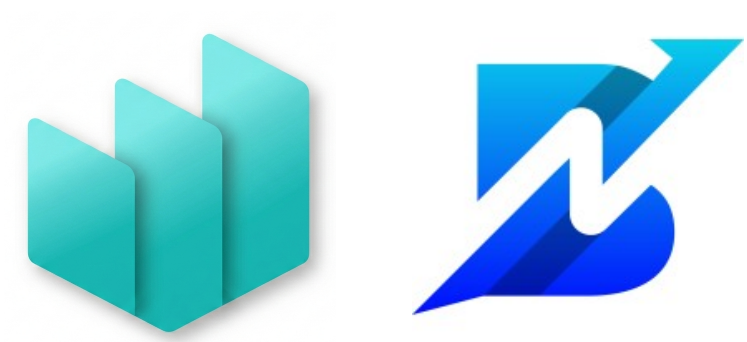




Long Only Alpha Modeling in Equity Markets

KRITI 2026

Documentation Report

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1 Introduction

Modern systematic trading requires a robust transition from signal generation to capital allocation. This project develops a Dual-Sleeve Hybrid Architecture tailored for the proprietary dataset, specifically engineered to navigate a high-friction environment where execution efficiency is paramount.

The strategy bifurcates capital into two uncorrelated engines: a high-alpha IPO Penny Stock Momentum Sleeve and a multi-factor Convex Policy Optimization Sleeve. At the core of the system is the formulation of the portfolio selection process as a Multi-Period Optimization (MPO) problem. While traditional models focus primarily on return maximization, our framework prioritizes volatility minimization and downside protection. By defining a convex objective function that strictly penalizes portfolio variance, risk-forecast errors, and transaction costs, the system identifies a risk-parity-aligned allocation that ensures stability under a self-financing condition.

This mathematical approach allows the strategy to systematically navigate the trade-off between alpha capture and a heavy transaction tax of 26.8 bps. Integrated with cross-sectional Z-scoring and a hysteresis-based turnover filter ensuring strict T+1 execution constraints.

2 Problem Formulation

The objective is to develop a fully systematic, rules-based, long-only equity trading strategy using historical daily data from the Indian equity market. The strategy must identify robust, scalable investment opportunities and adapt consistently across an in-sample development period (2010-2020) to maximize risk-adjusted returns on an unseen out-of-sample dataset (2020-2025). It must operate under realistic market constraints, including a fixed initial capital of 50,00,000, portfolio limits of 1 to 100 stocks, a strict prohibition on fractional shares, and the capability to allocate up to 100% in cash. Furthermore, the system must account for structural frictions by applying stringent transaction costs equivalent to 0.536% for a round-trip trade.

We framed the challenge not merely as a stock-picking exercise, but as a dynamic capital allocation problem requiring a balance between high-alpha momentum plays and stable, factor-driven systematic returns. To solve this, we engineered a dual-sleeve architecture combining an IPO Penny Stock Momentum strategy with a core multi-factor portfolio. We programmed dynamic risk-management rules to aggressively scale back the IPO allocation during severe drawdowns. For the core portfolio, we implemented a hysteresis-based feature selection model that ranks stocks based on engineered signals like trailing momentum, volatility, and liquidity. Instead of using naive equal-weighting, we fed these filtered signals into a convex optimization engine to mathematically maximize our return forecast while penalizing covariance risk and turnover.

3 Stock Selector

To construct a robust core portfolio capable of maintaining high risk-adjusted returns while absorbing structural frictions (such as the 0.536% round-trip transaction cost), we implemented a multi-factor ranking system. This Stock Selector evaluates the cross-section of the market on a daily basis, utilizing a custom composite indicator built from a blend of momentum, volatility, and liquidity metrics.

By applying cross-sectional normalization and a hysteresis-based regime filter, the selector isolates a dynamic universe of high-quality equities for the downstream convex optimization engine.

3.1 Feature Engineering and Mathematical Formulation

The raw daily market data is first transformed into a set of foundational features. For a given asset i at time t , with closing price $P_{i,t}$ and volume $V_{i,t}$, we define the daily logarithmic return as $R_{i,t} = \ln(P_{i,t}/P_{i,t-1})$. From this, we engineer six distinct components:

- **Long-Term & Medium-Term Momentum:** We capture the multi-horizon trend strength using 60-day and 20-day percentage price changes.

$$Mom_{60} = \frac{P_{i,t}}{P_{i,t-60}} - 1 \quad , \quad Mom_{20} = \frac{P_{i,t}}{P_{i,t-20}} - 1$$

- **Annualized Volatility:** We penalize erratic assets by calculating the 60-day rolling standard deviation of daily log returns, annualized assuming 252 trading days.

$$\sigma_{60} = \sqrt{252} \times \text{std}(R_{i,t-59:t})$$

- **Liquidity:** To ensure scalability and minimize slippage, we evaluate the 20-day simple moving average of traded volume.

$$Liq_{20} = \frac{1}{20} \sum_{k=0}^{19} V_{i,t-k}$$

- **Normalized Average True Range (NATR):** We utilize a 14-day rolling mean of the True Range (TR), normalized by the current closing price, to assess absolute price action volatility.

$$TR_{i,t} = \max(H_{i,t} - L_{i,t}, |H_{i,t} - P_{i,t-1}|, |L_{i,t} - P_{i,t-1}|)$$

$$NATR_{14} = \left(\frac{\frac{1}{14} \sum_{k=0}^{13} TR_{i,t-k}}{P_{i,t}} \right) \times 100$$

- **Rolling Beta:** We estimate systematic risk via a 60-day rolling covariance against the cross-sectional mean market return ($R_{M,t}$).

$$\beta_{60} = \frac{\text{Cov}(R_{i,t-59:t}, R_{M,t-59:t})}{\text{Var}(R_{M,t-59:t})}$$

3.2 Custom Composite Indicator and Normalization

Because these features exist on vastly different scales (e.g., millions for liquidity versus decimals for momentum), they cannot be combined directly. To solve this, we cross-sectionally normalize each feature daily into a standard Z-score. To prevent extreme outliers from skewing the model, the Z-scores are symmetrically clipped at $[-3, 3]$:

$$Z_{f,i,t} = \max \left(-3, \min \left(3, \frac{f_{i,t} - \mu_{f,t}}{\sigma_{f,t}} \right) \right)$$

Where $\mu_{f,t}$ and $\sigma_{f,t}$ are the cross-sectional mean and standard deviation of feature f across all active stocks on day t .

These standardized features are then linearly combined into our **Custom Composite Score** ($S_{i,t}$). The weighting schema is explicitly designed to favor strong momentum and liquidity while heavily penalizing high volatility:

$$S_{i,t} = (0.30 \cdot Z_{Mom60}) + (0.20 \cdot Z_{Mom20}) - (0.30 \cdot Z_{\sigma60}) + (0.10 \cdot Z_{Liq}) + (0.05 \cdot Z_{NATR}) + (0.05 \cdot Z_{\beta60})$$

3.3 Hysteresis-Based Turnover Control

A naive application of selecting the top N ranked stocks daily would result in severe turnover at the boundary, destroying returns through transaction costs. To mitigate this, we implement a **Hysteresis Filter**.

We rank the custom composite scores cross-sectionally in descending order. Instead of a hard cut-off, we define an entry threshold ($N_{enter} = 65$) and a wider exit threshold ($N_{exit} = 79$). Setting the maximum exit threshold at 79 guarantees that, when combined with the capacity of the IPO Sleeve, the combined portfolio strictly adheres to the competition’s 100-stock maximum constraint while maintaining broad cross-sectional diversification.

- **Entry:** A stock enters the eligible universe if its rank ≤ 65 .
- **Hold:** A stock remains in the universe as long as its rank < 79 .
- **Exit:** A stock is strictly removed once its rank ≥ 79 .

This dual-threshold logic creates a buffer zone that significantly reduces portfolio churn while ensuring we maintain exposure to high-ranking assets. By enforcing this memory-based state transition, the strategy organically limits the transaction costs that would otherwise erode the alpha generated by the composite signal.

4 Experimented Approach

4.1 Alternating Direction Method of Multipliers (ADMM)

To efficiently solve the NP-hard portfolio optimization problem during weekly rebalancing, we implemented the Alternating Direction Method of Multipliers (ADMM). ADMM decouples the quadratic objective from the non-linear budget constraint via alternating updates.

We define the augmented primal variable $x \in \mathbb{R}^{3n}$ by concatenating target weights w and positive/negative trading vectors $\Delta w^\pm$:

$$x = [w^T, (\Delta w^-)^T, (\Delta w^+)^T]^T$$

To isolate the budget constraint $\mathcal{C}$, we introduce an auxiliary consensus variable $y \in \mathbb{R}^{3n}$ and an indicator function $g(y)$ (where $g(y) = 0$ if $y \in \mathcal{C}$, and ∞ otherwise). Letting $f(x)$ encapsulate the quadratic objective and linear bounds, the problem becomes:

$$\min_{x,y} f(x) + g(y) \quad \text{subject to} \quad x - y = 0$$

We then form the Scaled Augmented **Lagrangian**, utilizing a scaled dual variable u and a penalty parameter $\rho = 1.0$:

$$\mathcal{L}_\rho(x, y, u) = f(x) + g(y) + \frac{\rho}{2} \|x - y + u\|_2^2$$

The ADMM algorithm minimizes this **Lagrangian** by iterating three steps until the primal residual converges below a 10^{-3} tolerance:

- **1. The x -update (Quadratic Programming Step):** Minimizing $\mathcal{L}_\rho$ w.r.t. x (holding y, u constant) yields a standard QP problem, utilizing objective matrix Q and linear coefficient vector R :

$$x^{(k+1)} = \arg \min_x \left\{ \frac{1}{2} x^T Q x - x^T R + \frac{\rho}{2} \|x - y^{(k)} + u^{(k)}\|_2^2 \right\}$$

Subject to the inventory transition from previous weights w_{prev} :

$$w = w_{prev} + \Delta w^+ - \Delta w^-$$

And standard boundaries, capped at a maximum weight $w_{max} = 0.20$:

$$0 \leq w \leq w_{max}, \quad \Delta w^+ \geq 0, \quad \Delta w^- \geq 0$$

To cast this into standard solver format, $R = \gamma[\mu^T, -c_{lin}\mathbf{1}^T, -c_{lin}\mathbf{1}^T]^T$ and Q embeds the covariance Σ and $2\gamma c_{quad}I$, constructed using risk aversion γ , linear costs c_{lin} , and quadratic costs c_{quad} .

- **2. The y -update (Non-linear Budget Projection via KKT Conditions):** Holding x and u

fixed, the non-linear budget constraint is enforced by projecting the target vector $v = x^{(k+1)} + u^{(k)}$ onto $\mathcal{C}$:

$$y^{(k+1)} = \arg \min_y \left\{ g(y) + \frac{\rho}{2} \|x^{(k+1)} - y + u^{(k)}\|_2^2 \right\}$$

Using a Lagrange multiplier λ , the **Lagrangian** for this projection is:

$$\Lambda(y, \lambda) = \frac{1}{2} \|y - v\|_2^2 + \lambda \left(\sum_{i=1}^n y_{w,i} + c_{lin} \sum_{i=1}^n (y_{dp,i} + y_{dm,i}) + c_{quad} \sum_{i=1}^n (y_{dp,i}^2 + y_{dm,i}^2) - 1 \right)$$

Evaluating the Karush-Kuhn-Tucker (KKT) stationarity conditions derives closed-form equations for the projected variables:

$$w_{proj}(\lambda) = \text{clip}(v_w - \lambda, 0, w_{max})$$

$$\Delta w_{proj}^{\pm}(\lambda) = \max \left(\frac{v_{\Delta w^{\pm}} - \lambda c_{lin}}{1 + 2\lambda c_{quad}}, 0 \right)$$

The optimal λ^* is the root of the substituted budget equation:

$$\sum_{i=1}^n w_{proj}(\lambda) + c_{lin} \sum_{i=1}^n (\Delta w_{proj}^{-}(\lambda) + \Delta w_{proj}^{+}(\lambda)) + c_{quad} \sum_{i=1}^n ((\Delta w_{proj}^{-}(\lambda))^2 + (\Delta w_{proj}^{+}(\lambda))^2) - 1 = 0$$

Since this function strictly decreases with λ , we utilize Brent's method to rapidly find the deterministic root within $[-10.0, 10.0]$ to form $y^{(k+1)}$.

- **3. The u -update (Dual Variable Update):** The scaled dual variable is updated using the residuals from the primal split:

$$u^{(k+1)} = u^{(k)} + x^{(k+1)} - y^{(k+1)}$$

- **4. Convergence Diagnostics:** The loop terminates when the L_2 norm of the primal residual diagnostic r falls below the predefined tolerance:

$$\|r^{(k+1)}\|_2 = \|x^{(k+1)} - y^{(k+1)}\|_2 < \text{tol}$$

Despite the theoretical efficiency of the ADMM framework in penalizing trades, its rigid non-linear penalty inadvertently forced the optimizer to hold onto deteriorating assets for too long during market corrections. By mechanically delaying necessary exits until the negative alpha finally eclipsed the strict transaction cost threshold, the strategy amplified tail risks and suffered a high maximum drawdown.

5 Convex Portfolio Optimization

At each node t , defined on a probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t=0}^T, \mathbb{P})$, the allocation engine maps the conditional expectation of returns

$$\hat{\boldsymbol{\mu}}_t = \mathbb{E}[\mathbf{r}_{t+1} \mid \mathcal{F}_t] \in \mathbb{R}^N$$

to a target weight vector $\mathbf{w}_t^* \in \mathbb{R}^N$. The random return vector $\mathbf{r}_{t+1}$ is assumed $\mathcal{F}_{t+1}$ -measurable with finite second moment,

$$\mathbb{E}[\|\mathbf{r}_{t+1}\|_2^2] < \infty.$$

Throughout, portfolio construction is treated as an optimization problem in which uncertainty in both first and second moments is explicitly incorporated. The guiding hypothesis can be stated formally as follows.

Let $\Omega_t \subset \mathbb{R}^N$ denote the admissible set of portfolio weights at time t . Portfolio construction is the problem of selecting

$$\mathbf{w}_t^* = \arg \max_{\mathbf{w} \in \Omega_t} \mathcal{J}_t(\mathbf{w}),$$

where $\mathcal{J}_t$ aggregates expected return, statistical risk, and robustness penalties derived from moment uncertainty.

The mathematical structure of $\mathcal{J}_t$ arises from a distributionally robust formulation: instead of assuming that the conditional distribution of returns is known, we allow it to vary within moment-based uncertainty sets and optimize against the worst admissible realization.

5.1 Uncertainty Sets and Regularization Penalties

We model deviations in conditional mean and covariance separately.

- **First-Moment Forecast Error:**

Assume the return admits the decomposition

$$\mathbf{r}_{t+1} = \hat{\boldsymbol{\mu}}_t + \boldsymbol{\delta}_t + \boldsymbol{\varepsilon}_{t+1},$$

where $\boldsymbol{\varepsilon}_{t+1}$ satisfies $\mathbb{E}[\boldsymbol{\varepsilon}_{t+1} \mid \mathcal{F}_t] = 0$, and $\boldsymbol{\delta}_t$ represents adversarial forecast bias.

We assume

$$\boldsymbol{\delta}_t \in \mathcal{U}_\mu = \{\boldsymbol{\delta} \in \mathbb{R}^N \mid -\boldsymbol{\rho}_t \preceq \boldsymbol{\delta} \preceq \boldsymbol{\rho}_t\}.$$

The worst-case degradation of expected return is then

$$\mathcal{E}_\mu(\mathbf{w}_t) = \sup_{\boldsymbol{\delta} \in \mathcal{U}_\mu} \boldsymbol{\delta}^T \mathbf{w}_t = \sum_{i=1}^N \rho_{t,i} |w_{t,i}| = \|\boldsymbol{\rho}_t \odot \mathbf{w}_t\|_1. \quad (1)$$

This quantity is the support function of $\mathcal{U}_\mu$ and appears naturally from a min-max formulation.

- **Second-Moment Model Uncertainty:**

Let the conditional covariance matrix belong to a discrete family

$$\mathfrak{S}_t = \{\Sigma_{diag}, \Sigma_{factor}\}.$$

The diagonal model captures idiosyncratic risk,

$$\Sigma_{diag} = \text{diag}(\sigma_1^2, \dots, \sigma_N^2),$$

while the factor representation is

$$\Sigma_{factor} = \mathbf{F}_t \Sigma_{f,t} \mathbf{F}_t^T + \mathbf{D}_t,$$

where $\mathbf{F}_t \in \mathbb{R}^{N \times k}$, $\Sigma_{f,t} \in \mathbb{S}_+^k$, and $\mathbf{D}_t \succeq 0$ diagonal.

The associated worst-case quadratic risk is

$$\Psi_t(\mathbf{w}_t) = \sup_{\Sigma \in \mathfrak{S}_t} \mathbf{w}_t^T \Sigma \mathbf{w}_t = \max \left(\mathbf{w}_t^T \Sigma_{diag} \mathbf{w}_t, \mathbf{w}_t^T \Sigma_{factor} \mathbf{w}_t \right). \quad (2)$$

- **Covariance Forecast Error:**

To model volatility instability, introduce

$$\Lambda_t = \text{diag}(\sigma_1, \dots, \sigma_N).$$

We penalize concentrated exposure through

$$\mathcal{E}_\Sigma(\mathbf{w}_t) = \|\Lambda_t \mathbf{w}_t\|_1^2 = \left(\sum_{i=1}^N \sigma_i |w_{t,i}| \right)^2. \quad (3)$$

This term behaves as a second-order regularizer controlling sensitivity to diagonal perturbations of covariance.

5.2 The Exact Formulation

Introduce parameters

$$\alpha > 0, \quad \lambda > 0, \quad \gamma > 0, \quad \eta > 0,$$

representing return scaling, first-moment robustness, overall risk aversion, and covariance perturbation weight respectively.

The allocation is determined by

$$\mathbf{w}_t^* = \arg \max_{\mathbf{w}_t \in \Omega_t} \left\{ \alpha \hat{\boldsymbol{\mu}}_t^T \mathbf{w}_t - \lambda \mathcal{E}_\mu(\mathbf{w}_t) - \gamma [\Psi_t(\mathbf{w}_t) + \eta \mathcal{E}_\Sigma(\mathbf{w}_t)] \right\}. \quad (4)$$

The objective may be interpreted as maximizing worst-case conditional expected return penalized by worst-case quadratic dispersion and structured volatility uncertainty.

Under positive definiteness of at least one admissible covariance model, the quadratic component induces strict curvature in the objective, ensuring uniqueness of the maximizer.

After extensive empirical examination and calibration, we have selected specific numerical values for $(\alpha, \lambda, \gamma, \eta)$ which are as $(1, 0.05, 15, 0.01)$.

5.3 The Polyhedral Feasible Space (Ω_t)

The admissible set is defined as

$$\Omega_t = \{ \mathbf{w} \in \mathbb{R}^N \mid \mathbf{w} \succeq \mathbf{0}, \quad \|\mathbf{w}\|_1 \leq 1, \quad \mathbf{w} \preceq \mathbf{m}_t \}. \quad (5)$$

This represents the intersection of the non-negative orthant, the L_1 unit half-space, and a coordinate-wise upper bound.

Compactness follows from boundedness in L_1 and closedness of the defining inequalities, which guarantees existence of an optimizer.

6 Multi-Period Extension (Horizon = 2)

In the single-period setting, the portfolio at time t is chosen by optimizing only the next period's trade-off between expected return and risk. In practice, however, trading decisions today influence both future portfolio composition and future transaction costs. A multi-period formulation accounts for this intertemporal dependence.

Let $\mathbf{w}_t$ denote the portfolio weights at time t . After trades are executed and returns are realized, the weights evolve according to

$$\mathbf{w}_{t+1} = \frac{(1 + \mathbf{r}_{t+1}) \odot \mathbf{w}_t}{1 + \mathbf{r}_{t+1}^T \mathbf{w}_t}. \quad (6)$$

This normalization ensures that weights sum to one after price changes. The update shows that future portfolio composition depends not only on returns but also on the current allocation choice.

6.1 Intertemporal Coupling Through Trades

Let normalized trades be defined as

$$\mathbf{z}_t = \mathbf{w}_{t+1} - \mathbf{w}_t.$$

We model transaction costs as a fraction of portfolio value V_t :

$$\phi_t(\mathbf{z}) = \tau \|\mathbf{z}\|_1,$$

where $\tau > 0$ captures brokerage, slippage, and statutory charges.

Because transaction costs reduce capital available in subsequent periods, the decision at time t affects both:

- immediate expected return,
- future portfolio flexibility.

This creates a genuine multi-period optimization problem.

6.2 Two-Period Objective Structure

In a two-period horizon, the objective at time t incorporates:

1. current period performance,
2. expected next-period performance,
3. transaction costs incurred in both periods.

The optimization problem can be written as

$$\max_{\mathbf{w}_t, \mathbf{w}_{t+1}} \mathcal{J}_t(\mathbf{w}_t) + \mathbb{E}_t[\mathcal{J}_{t+1}(\mathbf{w}_{t+1})] - \phi_t(\mathbf{z}_t) - \mathbb{E}_t[\phi_{t+1}(\mathbf{z}_{t+1})] \quad (7)$$

$$\text{s.t. } \mathbf{w}_t, \mathbf{w}_{t+1} \in \Omega. \quad (8)$$

The first term evaluates immediate performance. The second term anticipates how today's allocation influences tomorrow's risk–return trade-off. The cost terms penalize excessive turnover across both periods.

Thus, the optimizer does not simply maximize next-period return; it balances immediate gains against future stability.

6.3 Receding-Horizon Interpretation

Solving a full dynamic program over many periods is generally infeasible in large universes due to dimensionality. Instead, a finite-horizon approximation is used.

At time t :

1. A two-period (or finite-horizon) optimization is solved.
2. Only the first-period trade $\mathbf{z}_t$ is implemented.
3. At time $t + 1$, the problem is re-solved using updated information.

7 Implementation Framework

This section explains how the theoretical portfolio construction model is implemented in practice.

7.1 Market Data Preparation

For each stock and each trading date, we construct a representative execution price defined as the arithmetic average of open, high, low, and close prices:

$$p_{t,i} = \frac{O_{t,i} + H_{t,i} + L_{t,i} + C_{t,i}}{4}.$$

This reduces sensitivity to intraday noise and avoids anchoring decisions to a single print (e.g., close-only bias).

From these prices we compute next-period returns:

$$r_{t+1,i} = \frac{p_{t+1,i} - p_{t,i}}{p_{t,i}}.$$

A synthetic cash asset is appended with zero return, allowing the optimizer to allocate capital to cash explicitly.

Trading value data is also collected for each stock and date. An asset is considered tradable at time t only if its traded value is strictly positive. This prevents allocation to illiquid or delisted securities.

7.2 Dynamic Universe Control

Rather than optimizing over the entire stock universe at every step, the feature-based selection mechanism mentioned above determines which stocks are eligible for allocation.

7.3 Objective Construction

At each decision date, the optimizer solves a multi-period allocation problem whose objective contains three components:

1. Expected return forecast.
2. Risk term (worst-case across covariance models).
3. Covariance forecast error penalty.

The expected return component uses internally estimated return forecasts based on mean of all available historical data.

Risk is evaluated using two covariance structures:

- A purely diagonal covariance (idiosyncratic risk).

- A factor model covariance (systematic + idiosyncratic).

The optimizer penalizes the worst risk among these models, ensuring that the portfolio does not rely excessively on a single covariance specification.

An additional penalty is applied to control instability in covariance forecasts. This discourages concentrated exposure in highly volatile 'fid's.

The numerical weights assigned to these components are calibration parameters selected after empirical validation. They control the aggressiveness of the allocation.

7.4 Constraint Structure

The following constraints are enforced at every decision date:

- **Leverage limit:** total gross exposure does not exceed 1.
- **Long-only condition:** no short positions, including cash.
- **Universe mask:** only selected assets may carry positive weight.

These constraints ensure that the portfolio remains fully funded, long-only, and aligned with the dynamically selected universe.

7.5 Multi-Period Optimization Policy

The policy is implemented as a receding-horizon optimizer:

1. At time t , the engine forecasts returns and risk.
2. It solves a multi-period optimization problem.
3. Only the first trade is executed.
4. The process repeats at the next decision date.

This structure allows the model to incorporate forward-looking considerations without solving a full dynamic programming problem.

Trading frequency is set to weekly. This reduces turnover, lowers transaction costs, and stabilizes covariance estimation.

7.6 Execution Methodology and Cash Recursion

The optimization engine produces continuous target weights $\mathbf{w}_t^*$ at decision time t . In practice, however, portfolios are implemented in integer shares and executed at observable market prices. We therefore define a discrete execution mapping that converts continuous allocations into tradable quantities.

Let V_t denote the portfolio net asset value at time t , and let $\bar{\mathbf{p}}_{t+1}$ denote the execution price vector used for trade settlement. Following a $T + 1$ execution convention, trades are implemented using next-period prices.

The target integer share vector is defined as

$$\mathbf{q}_t = \left\lfloor \frac{\mathbf{w}_t^* V_t}{\bar{\mathbf{p}}_{t+1}} \right\rfloor, \quad (9)$$

where the floor operator enforces whole-share constraints. This guarantees that:

- fractional ownership is prohibited,
- allocations remain fully implementable under discrete trading.

Define the trade increment

$$\Delta \mathbf{q}_t = \mathbf{q}_t - \mathbf{q}_{t-1}.$$

Transaction costs are modeled as proportional to traded value:

$$\Phi(\Delta \mathbf{q}_t) = \tau \|\Delta \mathbf{q}_t \odot \bar{\mathbf{p}}_{t+1}\|_1, \quad (10)$$

where τ represents the aggregate trading friction (brokerage, slippage, statutory charges). In our implementation, $\tau = 0.268\%$.

The resulting cash recursion becomes

$$X_{t+1} = \max(X_t - \Delta \mathbf{q}_t^T \bar{\mathbf{p}}_{t+1} - \Phi(\Delta \mathbf{q}_t), 0). \quad (11)$$

This recursion ensures that capital is reduced by both traded value and transaction dissipation, while preventing negative cash balances.

7.7 Computational Considerations

The stock universe contains over one thousand assets. A full covariance matrix at this scale is computationally heavy and can destabilize the solver.

To maintain tractability:

- A factor model covariance is used.
- Universe masking reduces effective dimensionality.
- Weekly rebalancing reduces the number of optimization calls.

Overall, the implementation translates the theoretical robust multi-period allocation problem into a numerically stable and executable trading architecture with realistic constraints, discrete trading, and transaction cost modeling.

8 Dynamic IPO Handling Architecture

8.1 Introduction

The IPO Momentum Sleeve is engineered to capture explosive idiosyncratic returns in the Indian equity market while maintaining strict risk boundaries. It operates as a priority-weighted state machine that differentiates between fresh listings, missed opportunities, and secondary breakout candidates.

8.2 Universe Filtering and Asset Stabilization

The Five-Day Stability Rule

Let $\mathcal{U}_t$ denote the aggregate universe. For any asset i , let $t_0(i)$ be its listing date. We define the candidate set $\mathcal{C}_t$ by filtering for assets that have survived the initial five-day listing chaos and still not yet entered NSE500 and has future potential:

$$\mathcal{C}_t = \{i \in \mathcal{U}_t \mid \text{age}_i(t) = 5 \text{ and } i \notin \text{NSE500}\}$$

This observation window ensures that institutional flipping has subsided and a genuine market equilibrium is established before capital is committed.

Price-Based Asset Selection

To isolate penny stocks with high potential for asymmetric returns, the strategy enforces a hard price ceiling of $P_{max} = 40$ INR:

$$\Phi(i, t) = \mathbb{I}\{P_{i,t} \leq P_{max}\}$$

This threshold targets low-unit-price securities exhibiting high price elasticity, where minimal absolute price movements result in significant percentage gains. By filtering for assets under 40 INR, the system exploits the lower liquidity and high retail engagement typical of the micro-cap segment. This selection optimizes for "convexity," capturing rapid momentum breakouts where upside potential disproportionately outweighs the initial capital risk.

8.3 Longitudinal Capacity Scaling

The system's slot capacity N_{slots} is a dynamic function of the backtest duration Y (in years), ensuring the strategy has the "time-to-pop" necessary for IPO maturation:

$$N_{slots}(Y) = \text{clip}(\lfloor 5 + Y \rfloor, 5, 20)$$

This ensures that in multi-year datasets, the strategy expands its breadth to capture multiple IPO cycles without over-diluting individual position.

8.4 Position Sizing and Capital Utilization

Intra-Sleeve Allocation and Recursive Top-Up

The target weight for a new entry is relative to empty slots in order to compound our investments while maintaining cash to invest in other IPOs coming down the line:

$$w_{init} = \frac{K_{IPO,t}}{N_{slots} - N_{occupied}}$$

We enforce a single-position cap $\sigma = 0.10$ relative to the **IPO Portfolio Value** ($V_{IPO,t}$). If w_{init} exceeds this threshold, a recursive reinvestment occurs:

$$w_{final,i} = \min(w_{init}, \sigma \cdot V_{IPO,t})$$

If a residual cash balance K_{resid} exists due to the σ cap, $\alpha = 90\%$ of this excess is redistributed proportionally across existing holdings that have not yet reached the 10% saturation limit, minimizing cash drag.

8.5 Dynamic Risk Control and Capital Preservation

To protect the total account from extreme volatility, the strategy employs a two-tier risk management system. This includes an asset-specific exit strategy and a sleeve-level capital reallocation protocol based on total drawdown.

Asset-Level Trailing Stop-Loss

The primary exit mechanism is a price-based trailing stop-loss that locks in profits while limiting downside. For any active position i , the system maintains a High-Water Mark (HWM) $M_{i,t}$, representing the peak price achieved since entry:

$$M_{i,t} = \max_{\tau \in [t_{entry}, t]} \{P_{i,\tau}\}$$

A liquidation signal S_{exit} is triggered if the price retraces by $\lambda = 20\%$ from its HWM:

$$S_{exit,t} = \mathbb{1}\{P_{i,t} \leq 0.80 \cdot M_{i,t}\}$$

Sleeve-Level Drawdown Intervention

The strategy monitors the cumulative performance of the IPO Sleeve to prevent large-scale capital erosion. If the sleeve experiences significant drawdowns, capital is systematically moved from the high-risk IPO strategy into the more stable Core Portfolio Optimization strategy:

- **30% Drawdown Threshold:** If the total value of the IPO sleeve falls 30% from its peak, the system liquidates 50% of all active IPO positions on the following trading day. This capital is immediately reallocated to the Core Portfolio Sleeve to reduce exposure.

- **40% Drawdown Threshold:** If the IPO sleeve drawdown reaches 40%, a "full-exit" protocol is triggered. The system liquidates all remaining IPO positions and halts new entries. All remaining cash is transferred to the Core Portfolio to preserve principal.

By shifting capital into the optimized portfolio during periods of high risk exposure, the engine ensures that losses in the IPO sleeve do not compromise the overall viability of the total fund.

8.6 The Hierarchical Priority Queue System

The system manages secondary liquidity through an elaborate multi-tiered polling system. When a position slot is vacated, the engine performs a hierarchical sweep across three distinct candidate pools to ensure optimal capital rotation.

Tier 1: The Incoming Queue ($\mathcal{Q}_{incoming}$)

The highest priority is assigned to assets at $t = t_0(i) + 5$. By prioritizing fresh listings, the strategy ensures it is present for the initial momentum breakout that follows the expiration of listing-day price bands.

Tier 2: The Missed Opportunity Queue ($\mathcal{Q}_{missed}$)

Assets that met all entry criteria but were discarded due to $N_{occupied} = N_{slots}$ are held in $\mathcal{Q}_{missed}$.

- **Persistence Check:** Unlike fresh IPOs, missed candidates are only re-evaluated if they continue to trade below P_{max} .
- **Last-In, First-Out (LIFO) Processing:** The system treats these as a backlog, filling slots chronologically to ensure no high-torque asset is ignored simply because it listed during a crowded market regime. It prioritizes latest IPOs since they have lost much less of their potential as compared to IPOs which came earlier.

Tier 3: The TSL-Hit Recovery Queue ($\mathcal{Q}_{hit}$)

Assets exited via the Trailing Stop-Loss are demoted to the lowest priority tier. Re-entry from this queue is the most technically demanding, requiring the asset to overcome a **Dual Hysteresis Filter**:

1. **Temporal Cooldown:** A mandatory $\Delta t \geq 5$ days must pass to ensure the sell-off was not a structural collapse.
2. **Breakout Verification:** $P_{i,t} > P_{exit} \cdot 1.10$. This 10% hurdle acts as a confirmation of a secondary momentum wave, effectively filtering out "dead cat bounces" or sideways consolidation.
3. **Trading Cap:** Total trade frequency is capped at $N_{trades} \leq 10$ per ticker to prevent the system from over-trading a stagnant asset in a high-friction environment.

9 Strategy Integration and Final Results

To balance the high-alpha potential of the IPO Momentum Sleeve with the robust stability of the Convex Portfolio Optimization Sleeve, the system dynamically allocates initial capital based on the breadth of the market universe. By default, the architecture enforces a 30%–70% capital split between the IPO and Core sleeves, respectively. However, if the active stock universe exceeds 1,200 equities on the inception date, the allocation is defensively adjusted to a 10%–90% split. This dynamic adjustment accounts for the fact that in a highly populated, fragmented market, the probability of isolating explosive IPO breakouts diminishes. By shifting capital weight toward the systematically optimized core portfolio, the strategy minimizes exposure to idiosyncratic noise and preserves overall capital efficiency. To adhere by the minimum 1 position constraint, if the number of shares on any particular day is 0 we buy 1 share of the cheapest asset on that day.

Over the 10-year in-sample period (2009–2020), the Dual-Sleeve Architecture achieved a 37.43% CAGR, decisively outperforming the Nifty 500’s 8.61%. Stringent convex optimization and dual trailing-stops constrained the Maximum Drawdown to –25.08%, shielding capital much better than the benchmark’s –31.82% decline.

Risk-adjusted performance is exceptionally strong, marked by a Sharpe Ratio of 1.947 and a Sortino Ratio of 2.301. Crucially, the strategy displays a highly asymmetric capture profile: it deliberately trades some market upside (52.9% up-capture) to severely restrict losses (19.6% down-capture). This yields a massive 2.69 Capture Ratio, driving consistent compounding and expanding alpha generation across all rolling horizons (Table 2), easily absorbing the friction of a 14.3× annualized turnover.

Table 1: Average Annualized Performance Summary

Metric	Strategy	Nifty 500
CAGR	37.43%	8.61%
Annualized Std Dev	17.39%	15.05%
Maximum Drawdown	-25.08%	-31.82%
Sharpe Ratio	1.947	0.633
Sortino Ratio	2.301	–
Information Ratio vs Nifty 500	1.274	–
Up-Capture vs Nifty 500	52.9%	–
Down-Capture vs Nifty 500	19.6%	–
Annualized Turnover	14.3x	–

Table 2: Rolling Outperformance vs Nifty 500

Window	Avg Outperformance	Worst Underperformance
1Y	33.3%	-0.7%
3Y	190.4%	77.7%
5Y	582.5%	307.2%